

Mathematics for Social Scientists

Part 2: Introduction to Analysis Derivatives

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Outline

Derivatives

Intro

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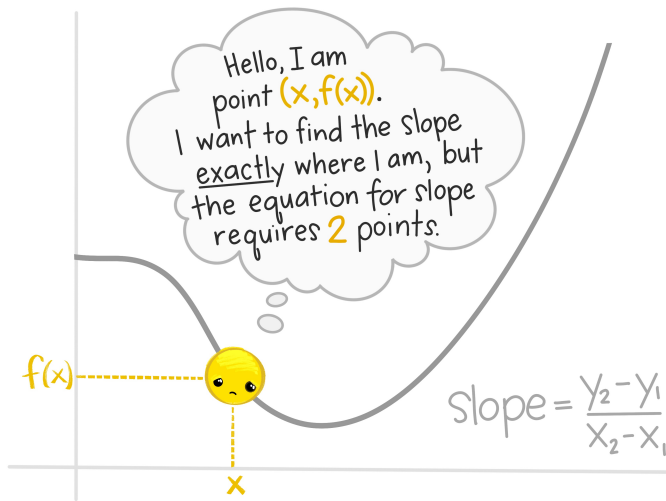
Differentiation Definitions

Derivative Rules

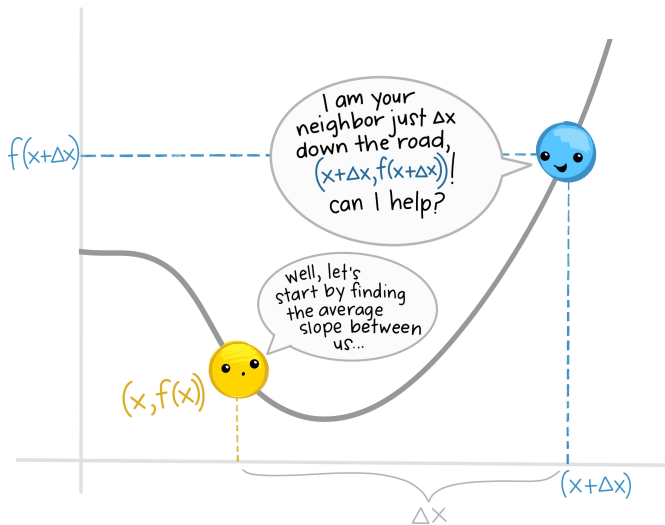
L'Hospital's Rule

Partial Derivatives

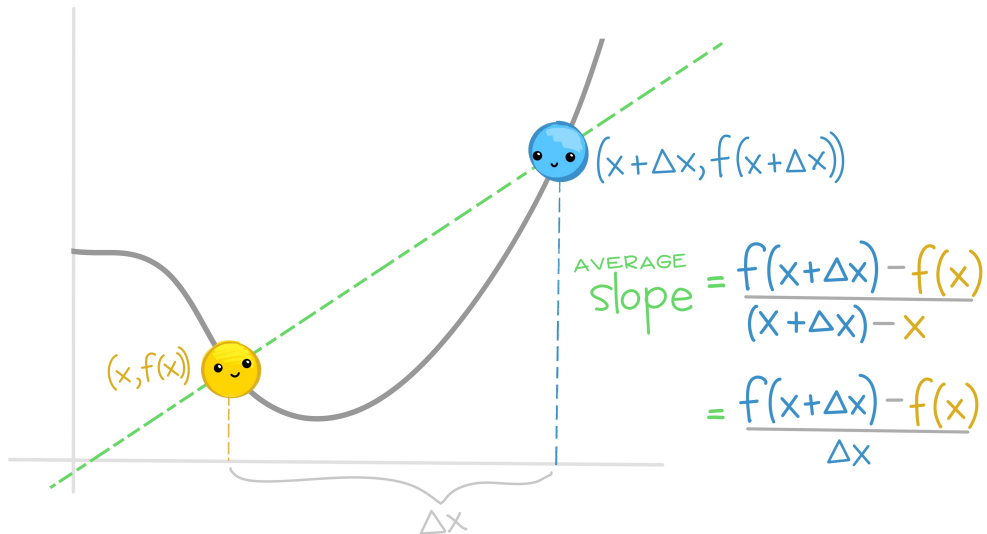
What is a derivative?



What is a derivative?



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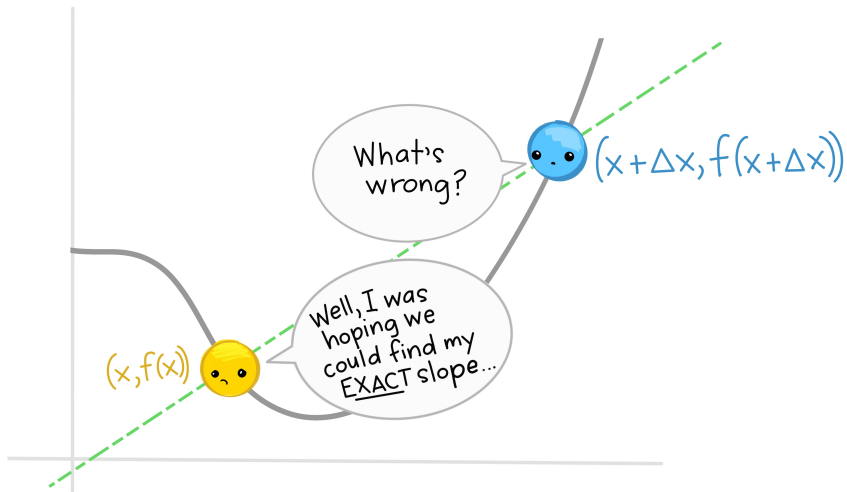


What is a derivative?

So: the average slope between
ANY 2 POINTS on function $f(x)$
separated by Δx is

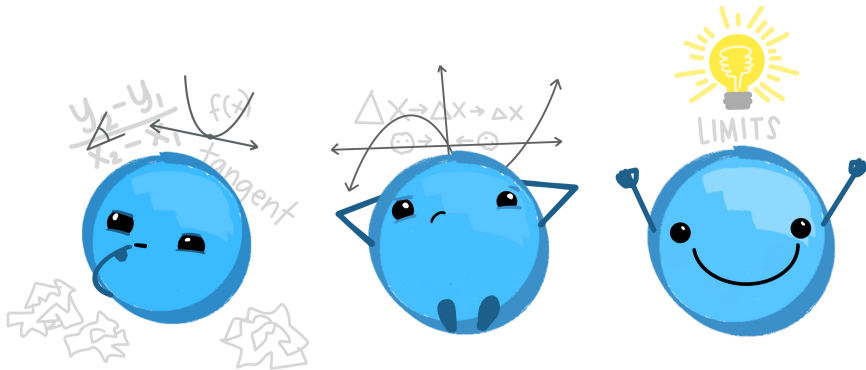
$$m = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

What is a derivative?

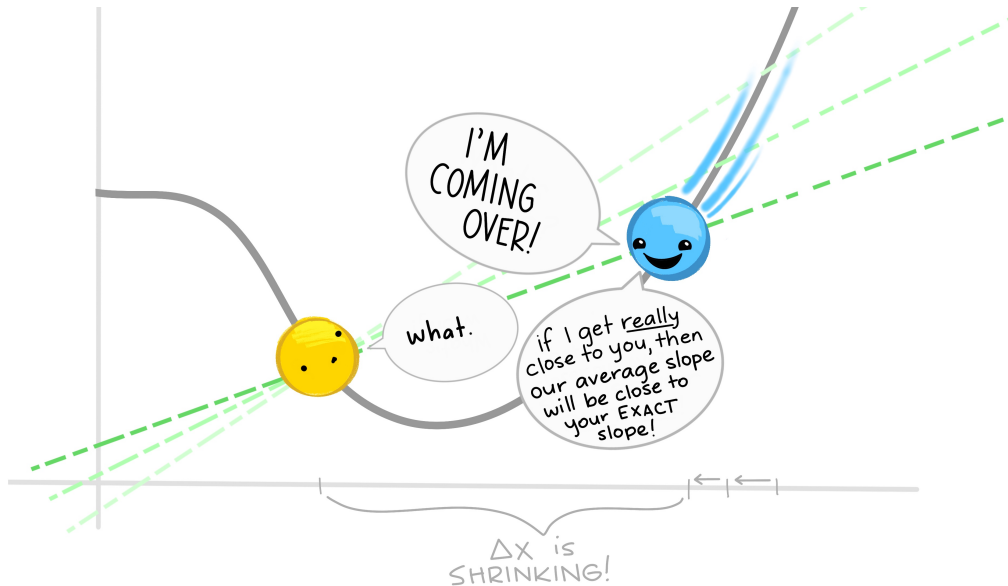


What is a derivative?

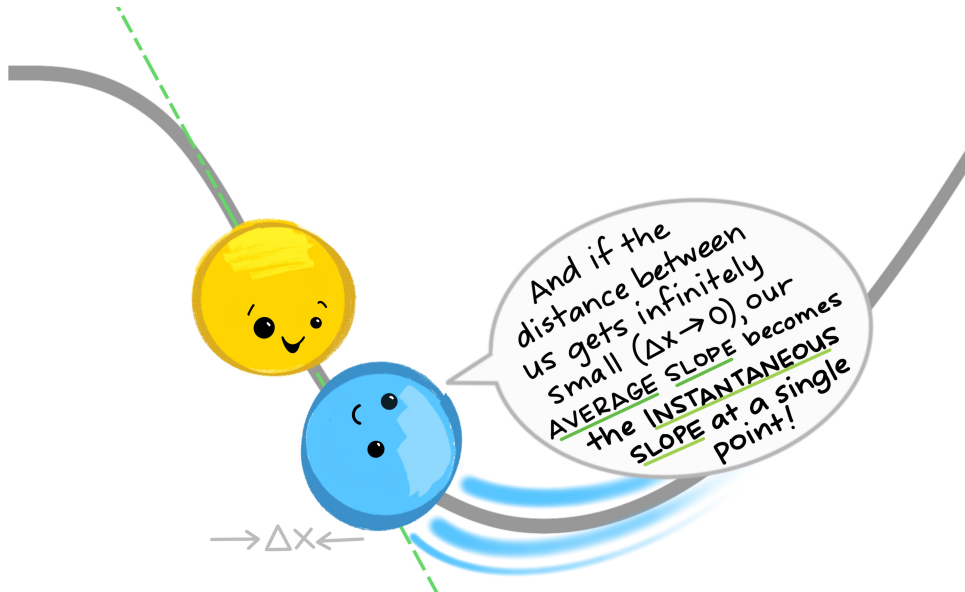
BRAINSTORM MONTAGE!



What is a derivative?



What is a derivative?



What is a derivative?

The expression for the instantaneous slope at any point on a function, aka the **derivative**



$$\frac{df}{dx} = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

IS FOUND BY:

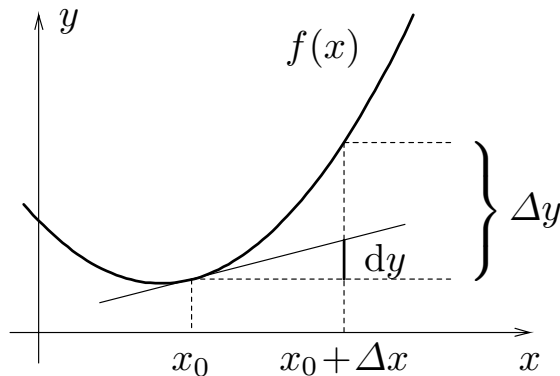
① Finding an expression for the **slope** between 2 points separated by Δx ...

② evaluating that slope as the points get infinitely close together.

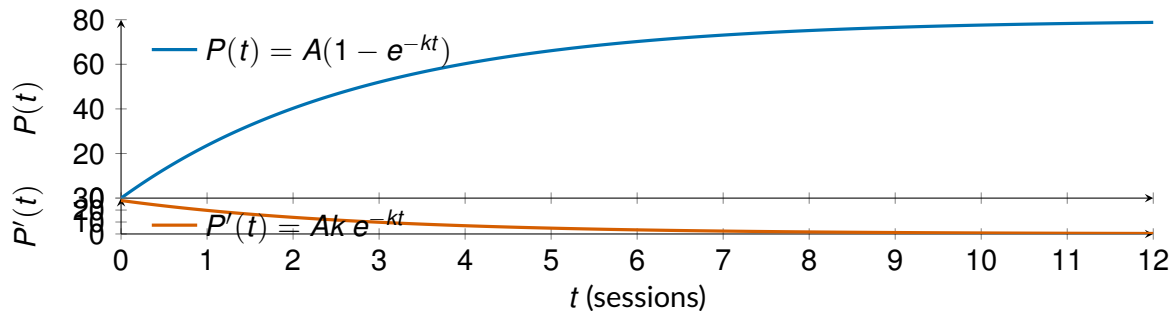
What is a derivative?

We want to estimate the slope of a function at point x_0 .

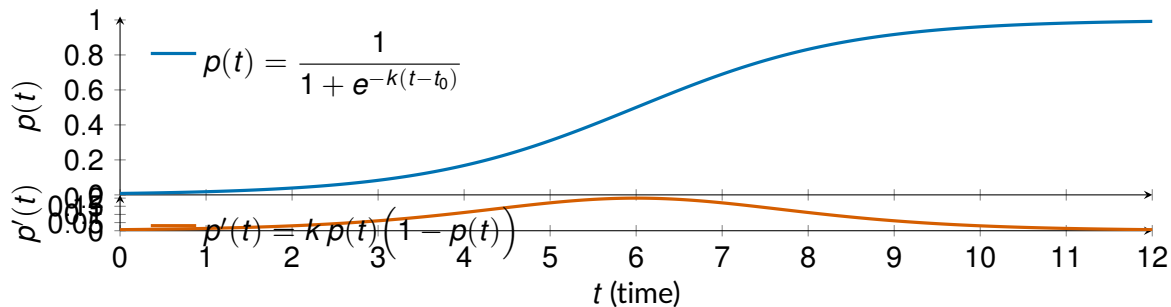
- Difference quotient $\frac{\Delta y}{\Delta x}$.
- Let $\Delta x \rightarrow 0$.
- Limit is the **derivative**.



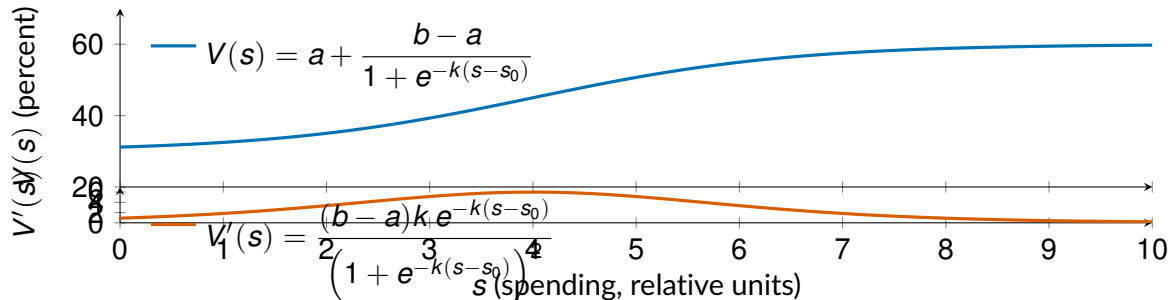
Psychology: learning curve and learning rate



Sociology: diffusion (logistic)



Political science: vote share vs. campaign spending

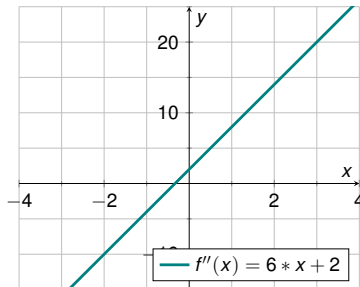
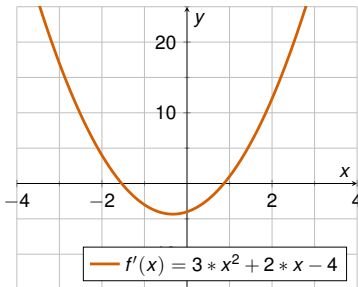
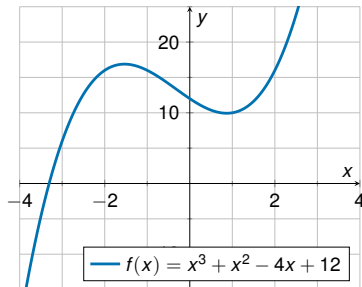


Differentiation Definitions

Rates, changes, and derivatives

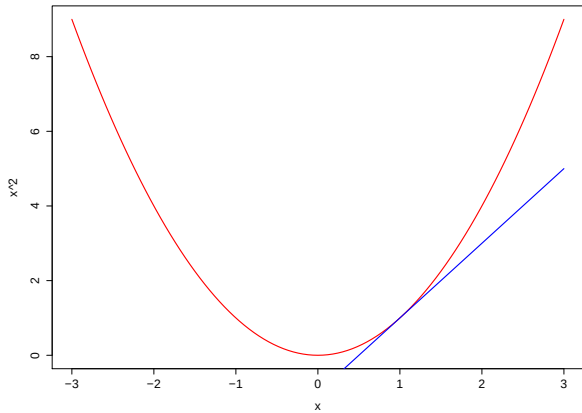
Our reason for studying limits is to learn about *derivatives*, which are defined by a limit.

Derivatives tell us about the rate of change of a function.



Tangents: Rise over run

General idea: Approximate a function locally by a linear function



Average vs. instantaneous rate of change

If $y = f(x)$ and x moves from x_1 to x_2 ,

$$\text{Average rate} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

Let $x_2 \rightarrow x_1$:

$$f'(x_1) = \lim_{x \rightarrow x_1} \frac{f(x) - f(x_1)}{x - x_1}$$

This is the slope of the *tangent line* at $(x_1, f(x_1))$.

Definition of differentiable functions

Definition 1.

Let $I \subseteq \mathbb{R}$ be an interval.

A function $f : I \rightarrow \mathbb{R}$ is called **differentiable at $a \in I$** if $\exists \varepsilon > 0$ such that $(a - \varepsilon, a + \varepsilon) \subseteq I$ and the limit

$$f'(a) := \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

exists.

This limit is called the **derivative at a** .

If f is differentiable at every $a \in I$, then the whole function is called **differentiable**.

(Alternative) Definition of differentiable functions

Note: differentiability at a point is a local property.

Alternative notation:

$$\frac{df}{dx} : I \rightarrow \mathbb{R} : y \mapsto f'(y)$$

Definition 2.

(Alternative Differentiation Definition) Another definition of the derivative of a function is given by the limit:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Q: Compute the derivative of $f(x) = 5x - \frac{1}{20}x^2$ using the alternative definition.

Q: Compute the derivative of $f(x) = x^2$ using the alternative definition.

Summary of Derivative Theory (J. Gill, 2006)

- **Existence:** $f'(x)$ at x exists $\iff f(x)$ is continuous at x , and there is no point where the right-hand derivative and the left hand derivative are different.
- **Definition:** $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
- **Tangent Line:** $f'(x)$ is the slope of the line tangent to $f()$ at x ; this is the limit of the enclosed secant lines.

Derivative Rules

Getting to practice: Important derivatives you should know

- $\frac{d}{dx} c = 0$ for any $c \in \mathbb{R}$
- $\frac{d}{dx} x = 1$
- $\frac{d}{dx} x^n = nx^{n-1}$, if $n \neq 0$
- $\frac{d}{dx} e^x = e^x$
- $\frac{d}{dx} \log(x) = x^{-1}$
- $\frac{d}{dx} \sin(x) = \cos(x)$
- $\frac{d}{dx} \cos(x) = -\sin(x)$

Relationship between differentiability and continuity

Proposition 1.

Let f be differentiable at a . Then f is also continuous at a .

Remark: The converse is not true in general:

Example.

$f(x) = |x|$ is continuous at 0 but not differentiable at 0.

Combinations of differentiable functions I

Proposition 2.

Let f, g be functions that are differentiable in a . Then, the functions

$$f + g, \quad c \cdot f \quad (c \in \mathbb{R})$$

are also differentiable in a , and

$$(f + g)'(a) = f'(a) + g'(a), \quad (cf)'(a) = cf'(a).$$

Combinations of differentiable functions II

Proposition 3 (The product rule).

Let $f, g : I \rightarrow \mathbb{R}$ be functions that are differentiable in a . Then, the functions

$$f \times g$$

is also differentiable in a , and

$$(f \times g)'(a) = f'(a)g(a) + f(a)g'(a).$$

Proposition 4 (The quotient rule).

If $g(x) \neq 0 \forall x \in I$ then the product rule can be used to show

$$\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{g^2(a)}.$$

The chain rule - differentiating compositions

Theorem 1 (Chain rule).

Let $f : I \rightarrow \mathbb{R}$, $g : J \rightarrow \mathbb{R}$ be two functions such that $f(I) \subseteq J$. Suppose that f is differentiable at a and that g is differentiable at $f(a)$. Then, the function $g \circ f$ is differentiable at a and

$$(g \circ f)'(a) = g'(f(a)) \cdot f'(a).$$

“The derivative of the inside, times the derivative of the outside *with respect to* the inside”.

Exercise: $f(x) = x^2$, $g(x) = \sin(x)$.

Derivatives of Higher Order

Taking the derivative of a function two times, we say we have a *derivative of second order* and denote it:

$$f''(x) = \frac{d^2 f(x)}{dx^2} = \frac{d}{dx} \left(\frac{df(x)}{dx} \right)$$

In general, the n^{th} order derivative is shown as:

$$f^{(n)}(x) = \frac{d^n f(x)}{dx^n}$$

We say that the *derivative is of order n* .

Ex: Find the second and third order derivatives of $f(x) = x\sin(x)$.

$f'(x) = \sin(x) + x\cos(x) \rightarrow$ 1st order derivative

$f''(x) = \cos(x) + \cos(x) - x\sin(x) = 2\cos(x) - x\sin(x) \rightarrow$ 2nd order derivative

$f^{(3)}(x) = -2\sin(x) - \sin(x) - x\cos(x) = -3\sin(x) - x\cos(x) \rightarrow$ 3rd order

L'Hospital's Rule

L'Hospital's Rule

L'Hospital's Rule is a way to evaluate special limits that are in the form of $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

Theorem 2 (L'Hospital's Rule).

Assume that $f(x)$ and $g(x)$ are differentiable functions at a , where $f(a) = 0$ and $g(a) = 0$ (or both equal to ∞).

Then:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)},$$

provided that $g'(x) \neq 0$.

Q: Evaluate the following limit: $\lim_{x \rightarrow 0} \frac{x}{\log(1-x)}$

Q: Evaluate the following limit: $\lim_{x \rightarrow \infty} \frac{(\log(x))^2}{x^2 \log(x)}$

Partial Derivatives

Multivariate Functions

Definition 3.

Let $D \subseteq \mathbb{R}^n$. A multivariate function is a map

$$f : D \rightarrow \mathbb{R}, \quad \mathbf{x} = (x_1, \dots, x_n) \mapsto f(\mathbf{x}).$$

The set D is the domain of f .

Example.

(i) $f(x, y) = x^2 + 3y$ has domain $D = \mathbb{R}^2$. (ii) $g(x, y) = \ln(x - y)$ has domain $D = \{(x, y) : x - y > 0\}$.

Limits and Continuity

Definition 4 (Limit).

Let $f : D \rightarrow \mathbb{R}$ and $\mathbf{a}$ be a limit point of D . We write

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = L$$

if $f(\mathbf{x})$ can be made arbitrarily close to L by taking $\mathbf{x}$ close to $\mathbf{a}$ (with $\mathbf{x} \in D$, $\mathbf{x} \neq \mathbf{a}$).

Definition 5 (Continuity).

f is continuous at $\mathbf{a} \in D$ if

$$\lim_{\mathbf{x} \rightarrow \mathbf{a}} f(\mathbf{x}) = f(\mathbf{a}).$$

Example.

$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$ does not exist: along $y = x$ it tends to $\frac{1}{2}$, along $y = 0$ it tends to 0.

Partial Derivatives: Definition

Definition 6.

Let $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}$ and $\mathbf{a} = (a_1, \dots, a_n) \in D$. For $i \in \{1, \dots, n\}$, the *partial derivative* of f with respect to x_i at $\mathbf{a}$ is

$$\frac{\partial f}{\partial x_i}(\mathbf{a}) = \lim_{h \rightarrow 0} \frac{f(a_1, \dots, a_i + h, \dots, a_n) - f(\mathbf{a})}{h},$$

provided the limit exists.

Notations: f_{x_i} , $\partial_{x_i} f$, $D_{x_i} f$, $\frac{\partial f}{\partial x_i}$ (all equivalent).

Rule of thumb: hold all variables other than x_i fixed and differentiate as in one variable.

Partial Derivatives: In simpler terms

Partial derivatives are regular derivatives, but where the function has multiple variables and you take the derivative with respect to one of these variables, treating other variables as if they were constants.

Example.

Take the derivative of $f(x, y) = 3x + 2y$ with respect to x .

$$f_x = \frac{\partial f(x, y)}{\partial x} = 3.$$

Partial Derivatives II

In general, let

$$u = f(x, y, z)$$

Then, the three partial derivatives are:

$$f_x(x, y, z) = \frac{\partial u}{\partial x}$$

$$f_y(x, y, z) = \frac{\partial u}{\partial y}$$

$$f_z(x, y, z) = \frac{\partial u}{\partial z}$$

Interpretations of Partial Derivatives

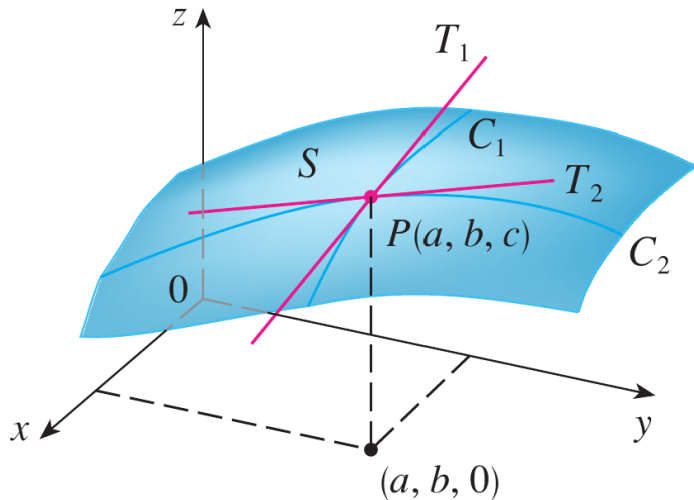


Figure: The partial derivatives of f at (a, b) are the slopes of the tangents to C_1 and C_2 .

The second derivative of a function $\mathbb{R}^d$

If the partial derivatives $\partial_1 f, \dots, \partial_d f$ are again partially differentiable, one can form the partial derivatives of the second order of f :

Notation: $\partial_{ij} f := \partial_i(\partial_j f)$.

Example.

$f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = x^3 \cdot y^2$ Then:

$$\partial_{xx} f(x, y) = 6xy^2 \quad \partial_{xy} f(x, y) = 6x^2y$$

$$\partial_{yx} f(x, y) = 6x^2y \quad \partial_{yy} f(x, y) = 2x^3$$

Note that $\partial_{xy} f = \partial_{yx} f$. There's a theorem for that!

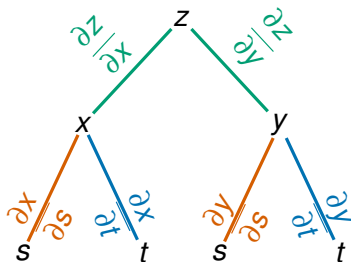
Schwarz' theorem

Theorem 3 (Schwarz).

Assume that the function f has the partial derivatives $\partial_i f$, $\partial_j f$ and $\partial_{ji} f$. Assume that $\partial_{ji} f$ is continuous in a . Then: $\partial_{ij} f(a)$ exists and

$$\partial_{ij} f(a) = \partial_{ji} f(a).$$

Multivariate Chain rule



Suppose $z = f(x, y)$, with $x = x(s, t)$ and $y = y(s, t)$ (all differentiable). Then

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

Sum over all paths from s (or t) up to z —multiply the derivatives along each path.

Appendix: Implicit differentiation in two variables

If $F(x, y) = 0$ defines y as a function of x near a point with $F_y \neq 0$, then

$$\frac{dy}{dx} = -\frac{F_x(x, y)}{F_y(x, y)}$$

Example.

Ellipse with tilt: $F(x, y) = x^2 + 4xy + 5y^2 - 1 = 0$.

$$F_x = 2x + 4y, \quad F_y = 4x + 10y.$$

$$\frac{dy}{dx} = -\frac{2x + 4y}{4x + 10y}.$$

At $(x, y) = (0, 1/\sqrt{5})$ the slope is $-\frac{4/\sqrt{5}}{10/\sqrt{5}} = -\frac{2}{5}$.

Handy when y isn't given explicitly.

Recommended Reading

Andrew Heiss

Published: May 20, 2022

Marginalia: A guide to marginal effects

DOI: 10.59350/40xaj-4e562

- What “marginal” means (as slopes/derivatives), and how to interpret **marginal effects**, **average marginal effects (AME)**, **marginal effects at the mean (MEM)**, and **marginal effects at representative values (MER)**.
- Hands-on comparison of `marginalEffects` and `emmeans` (R) and why the *order of averaging* changes results.
- <https://marginaleffects.com/>